Lecture 0: Overview of the Course

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Econometrics, Fall 2026

Plan for Today

Welcome to Econometrics!

Plan for Today:

- Introduction (Why do we study econometrics?)
- Overview of Syllabus and Class Policies
- Class Survey
- Start Stats Refresher (will finish Wednesday)

The Scientific Method in Economics

The scientific method is an iterative process:

- Develop theories
  - Scientific theories are *testable*, they have empirical implications
- Test those theories
  - Test them with data, either from experiments or observation

Many of your core classes in the major have focused mostly on theory

This class is all about *testing* and *quantifying* those theories

Social Science and Statistics

In the natural sciences, *often* do not need sophisticated statistical techniques:

- If I launch a smooth 10-pound steel sphere at 450 mph, it will follow (almost) the same trajectory each time

In the social sciences, we typically need statistics:

- If I give an 18-year-old a $5,000 college scholarship, it may make them more likely to attend college (Dynarski 2003), but attendance will vary across individuals

Why is social science so different?

- **Less Controlled Setting:** Can use identical cannonball; can't use identical people
- **Less Complete Theory:** Physics almost exactly describes cannonball's trajectory; economics makes prediction about averages

People vary in all sorts of ways and do things for all sorts of reasons

Econometrics vs. Other Branches of Statistics

Econometrics is especially focused on identifying causal relationships

- Return to Schooling: How much more would a worker earn if she went to college?

The gold standard is a randomized experiment

- We will discuss these extensively!

But, often not possible to run a randomized experiment

- People will typically not let you randomize whether they go to college

Without randomization, correlation need not imply causation

- Do college-educated workers earn more because of college, or because they were already different on average?

Econometricians have developed other techniques to estimate causal effects

- Focus on isolating *natural experiments* with as-good-as-random assignment

Linear Model

In this course, we will focus mainly on the linear model

Fits the following relationship to the data:

$$Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + ... + \beta_K X_{K,i} + \varepsilon_i$$

- Y_i is the outcome for observation i
- $X_{1,i}, ..., X_{K,i}$ are one or more regressors that affect the outcome
- α is an intercept, $\beta_1, \beta_2, ..., \beta_K$ are effects of the regressors on the outcome
- ε_i is an "error term" that captures everything else that affects the outcome

The linear model can actually be extended to fit nonlinear relationships, e.g. by including polynomial terms (X_i^2 and X_i^3) as regressors

Examples of the Linear Model

The linear model can capture many of the relationships we are interested in as economists

- **Labor Supply:** $\text{Hours Worked}_i = \alpha + \beta \cdot \text{Wage}_i + \varepsilon_i$
- **Consumer Demand:** $\text{Quantity Purchased}_i = \alpha + \beta \cdot \text{Price}_i + \gamma \cdot \text{Competitor Price}_i + \varepsilon_i$
- **Firm Production:** $\log(\text{Output}_i) = \alpha + \beta \cdot \log(\text{Capital}_i) + \gamma \cdot \log(\text{Labor}_i) + \varepsilon_i$
- **Returns to Education:** $\log(\text{Wage}_i) = \alpha + \beta \cdot \text{Years of Schooling}_i + \varepsilon_i$
- **Program Evaluation:** $Y_i = \alpha + \beta \cdot \text{Treated}_i + \varepsilon_i$
- ... and many, many more!

Goals of the Course

By the end of this course, you will be able to:

- Estimate and interpret linear models like the ones in the previous slide
- Test hypotheses and construct confidence intervals
- Use various techniques to estimate causal effects
- Understand the assumptions needed for each technique

More broadly, this course will prepare you to:

- Read empirical papers in the academic economics literature
- Understand the use, and misuse, of statistics in the popular press
- Start independent research using economic data, including in upper-level courses
- Analyze data in a variety of professional and academic settings

Class Logistics

Lecture on Mondays and Wednesdays with me

Tutorial on Fridays with your TAs, Bhavya Sinha and Tyler Boes

Office Hours:

- Majerovitz: Monday and Wednesday 5–6 PM in Jenkins Nanovic 3013
- Boes: Tuesday 1–2 PM in Jenkins Nanovic 3005
- Sinha: Thursday 2–3 PM in Jenkins Nanovic 3060

Peer Tutoring:

- Monday–Wednesday 6:30–8:30 PM in 240 DeBartolo Hall (Starts August 31)
- See more details at https://economics.nd.edu/undergraduate/peer-tutoring/

Final Exam: Thursday, December 17, 10:30 AM–12:30 PM

Prerequisites

We will assume:

- Good grasp of the concepts in ECON 30340 (Statistics for Economics)
- Basic knowledge of calculus

If you have not been exposed to this material before, please take ECON 30340 or an equivalent course before taking this class

We will briskly review this material this week, and the first problem set will cover it

- This material is essential, and we will use it over and over throughout the semester
- If you are rusty, I **strongly recommend** that you put in some extra time now to get comfortable with this material again: it will pay dividends for the whole semester

Course Requirements and Grading

Your grade will be determined by three components:

- **Problem Sets (35%):** Must be turned in on time. Graded as check-minus (60%), check (80%), or check-plus (100%). Most problem sets will get a check-plus.
- **In-Class Quizzes (30%):** Four closed-book, in-class quizzes. Seventy minutes each. I will drop your lowest quiz.
- **Final Exam (35%):** Closed-book exam, covering material from entire course.

Your grade will depend solely on the above formula. There will be no extra credit.

Course Policies

- **No laptops, no tablets, and no phones in lecture.**
  - I will distribute printed lecture notes for each class. I encourage you to also bring a small notebook for additional notes and to solve practice problems.
- You may bring a laptop to tutorial for Stata sessions.
- If you require disability accommodations, you must register with Sara Bea Accessibility Services. I encourage you to do this as soon as possible. University policy prohibits any accommodation except through that process.
- Upon entering Notre Dame you were required to sign a pledge to abide by the University's Honor Code. Students will not give or receive aid on exams. This includes, but is not limited to, viewing the exams of others, sharing answers with others, and making unauthorized use of books or notes while taking the exam.
- Although you may use AI tools to help you study, you may not use AI to complete your assignments.

More Course Policies

- There are five acceptable reasons for missing a quiz, exam, or problem set (see syllabus for details). I will accept no other reasons for missed work.
- If you miss an assessment for an excused reason, I will provide you the opportunity to make up the work, and/or reweight your other assessments appropriately.
- You must take the final exam at the designated time, unless you have an excused absence or are eligible for a reschedule due to a conflict with another final exam.
- Regrade requests must be received within two weeks of your quiz/problem set being returned. Regrade requests trigger a full regrade, so your grade could go up or down.

See syllabus for further details on course policies.

Stata

You will practice data analysis with Stata in some of the problem sets

You will learn how to use Stata in tutorials

We have obtained a free six-month license for members of this class to use I will send out an email with details of how to download Stata and access the free license

Course Schedule

Course divided into four parts, with each quiz covering one part:

1. Statistics Fundamentals and Ordinary Least Squares
2. Standard Errors, Confidence Intervals, and Hypothesis Tests
3. Instrumental Variables
4. Additional Methods (Regression Discontinuity, Panel Data, and Difference-in-Differences)

Second half of the course will incorporate "Econometrics in the Wild" lectures

- Apply the tools we have learned to recent research

See syllabus for full schedule

A Few Notes on What to Expect

This is a math class

- We will build the tools of statistical analysis on the chalkboard
- We will make precise mathematical statements about their properties and the assumptions they require
- I will not expect you to memorize *proofs*, but you should know key results

This is also an economics class

- I want you to understand the intuition as well as the math
- We will incorporate examples throughout, especially in the second half

This is a challenging class

- Material is cumulative: important not to fall behind
- Consistent effort throughout the semester will pay off

Class Survey and Stats Refresher

First, please fill out the class survey

- All questions are optional: this is just to help me get to know all of you better

Once that's done, we'll get started on our review of statistics

For that we will move to the chalkboard...